

Week 3 Report

Implementing Version 2 of the App with More Features

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1 Team & Roles

Team name: Fitness Challenge App with Camera-Based Exercise Verification

Repository: <https://github.com/Inno-Fitness/Fitness-Challenge-App>

Name	Email	Responsibilities
Ivan	i.lokhmotov@innopolis.university	Product coordination, setting up the server, Github page management view
Kristina	kr.ushakova@innopolis.university	Registration logic change, design CV and challenge pages view
Elvina	e.belorusova@innopolis.university	Improved the pushup counter accuracy, CV integration view
Daria	d.komzolova@innopolis.university	Setting up the DB and API tests creation view1 , view2
Safina	s.mukhamedshina@innopolis.university	New data addition, server deploying through docker view
Ekaterina	e.salazkina@innopolis.university	Integration of challenges into the website view1 , view2
Anastasia	a.kalashnikova@innopolis.university	Figma designs for the challenges screen (mobile and PC versions)
Timur	t.sagdeev@innopolis.university	Weekly deliverables (will be uploaded on Github later)

1.1 Chosen Project and Track

Project: WowFit Challenges

Track: Industrial

1.2 Week 3 Goal

Build on the MVP and implement the next-priority features. In this week we focused on the backend for registration and challenges, connected the data layer, updated the frontend design, and added tests and documentation for the new version.

2 Project Plan

2.1 What was planned for Version 2

- Extend registration and challenge functionality on top of the working MVP.
- Add backend support for users and challenges.
- Update the frontend design for the new flows.

- Add more seed/demo data.
- Write comprehensive API tests and update documentation.

3 Requirements Artifacts

3.1 How V2 Extends the MVP

- **Challenge implementation on the backend:** we implemented the full challenge logic. In the app, a *challenge* is a special training task created by a user for one exercise and one goal. For example, a user can create a challenge like “do 10 pushups” and share it with other participants. After that, the participant can open the challenge and complete it by doing the exercise on camera. This makes the app more than a simple workout tracker, because it adds a concrete goal, a shared activity, and a way to track completion.

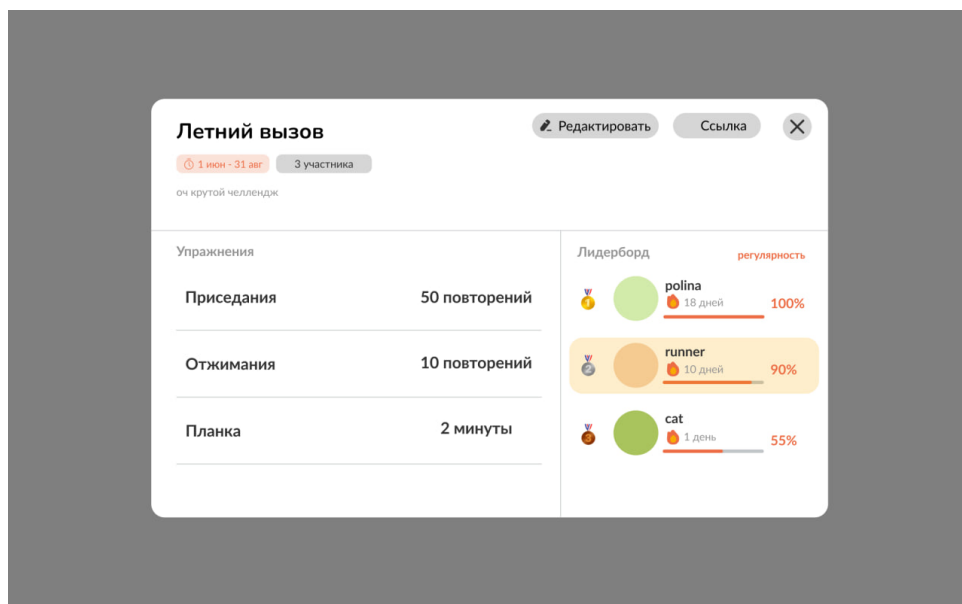


Figure 1: Checking the current challenge

- **Registration update:** the registration flow now requires `username`. This makes users easier to identify inside the app. At the same time, `firstname` and `lastname` became optional, so the sign-up process is simpler and more flexible.

Figure 2: Registration tab, note that mandatory fields are marked with *

- **Database support for users and challenges:** we added PostgreSQL support for storing users and challenges. This means the application now keeps the main product data in a structured database instead of relying only on temporary state, which makes the app more stable and usable across sessions.
- **Pushups data added:** we added pushups data to support the exercise content of the app. This could be used for onboarding in the future and to show the correct technique of doing the exercise.
- **Frontend design changes:** the design of the app was updated to match the new version. The interface now corresponds to the expanded feature set.

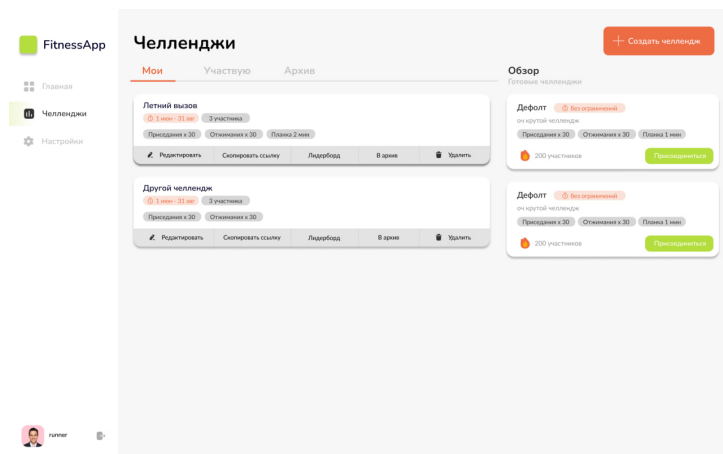


Figure 3: Challenges title page

- **Comprehensive API tests and documentation:** we added tests for the challenges API and updated the documentation. This helps verify that the new functionality works correctly and that the backend behavior is described clearly for future development.

3.2 Backend

- The full backend logic for challenges was implemented.

- The registration flow was updated to require `username`, `firstname` and `lastname` were made optional.
- PostgreSQL support for users and challenges was added.
- Pushups data was added for the exercise content and demo flow.

3.3 Frontend

- The frontend design was updated to reflect the richer product version.
- The new screens and flows were adjusted to support challenge creation and completion.
- The registration UI was changed to match the new username-based model.

3.4 Integration & Data

- New frontend features were connected to backend APIs.
- Database schema changes were applied for users and challenges.
- Validation rules were strengthened for registration and challenge-related data.

3.5 Tests & Documentation

- Comprehensive API and database tests for challenges were added.
- The documentation was updated to describe the new and changed backend behavior.
- The API docs now match the implemented functionality more closely.

4 Track-Specific Development

4.1 Industrial Track

Expanded contribution: the team expanded the existing product by finishing the backend challenge and registration implementation, strengthening the database layer, and improving the frontend presentation.

Updated measurements against the baseline:

Metric	Baseline	Week 3 result
Backend coverage	MVP only	Full challenge and registration logic implemented
Database support	Users only	Users and challenges DB added and connected
Validation	Basic registration fields	Username required; first and last name optional
Testing	Limited API coverage	Comprehensive API tests for challenges added

Demo data	Minimal	Push ups data added
UI / design	Simple MVP layout	Updated design applied

Compared with the MVP baseline, Week 3 adds a significantly more complete backend, a cleaner and more user-oriented registration flow, a proper data model for users and challenges, better API coverage through tests, and a refreshed frontend design.

4.2 Demo Summary

The expanded version was demoed to the team. The main backend flows worked, registration used the new username-based structure, and the frontend reflected the updated design. Challenges are creating correctly and overall the product began to look broader and more user-encouraging.

4.3 Updated Backlog

Finalize current features polish.

Add any missing tests.

Improve UI consistency and overall performance / accuracy.

Prepare for showing the MVP and for potential improvements in the future weeks.

5 Appendix: Useful Links

- Repository: <https://github.com/Inno-Fitness/Fitness-Challenge-App>
- Issues / backlog: <https://github.com/Inno-Fitness/Fitness-Challenge-App/issues>
- Demo video: https://drive.google.com/file/d/1Y8vRFv3_OvwLtjjIPDtBfJ0vUyqz1PhV/view?usp=sharing
- Deployed app: <http://10.93.27.12>